

Capture the Flag 3: Data-Driven UI

Goal: Separate application data from the UI and handle multiple pieces of data in a usable interface.

Instructor Note

These activities require your own problem-solving and research skills.

DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- **Explain** an error (not **fix** your error)
- Clarify documentation

You may **NOT** ask AI to generate the solution/code for a flag.

Flag 0 — Create Your Feature Branch

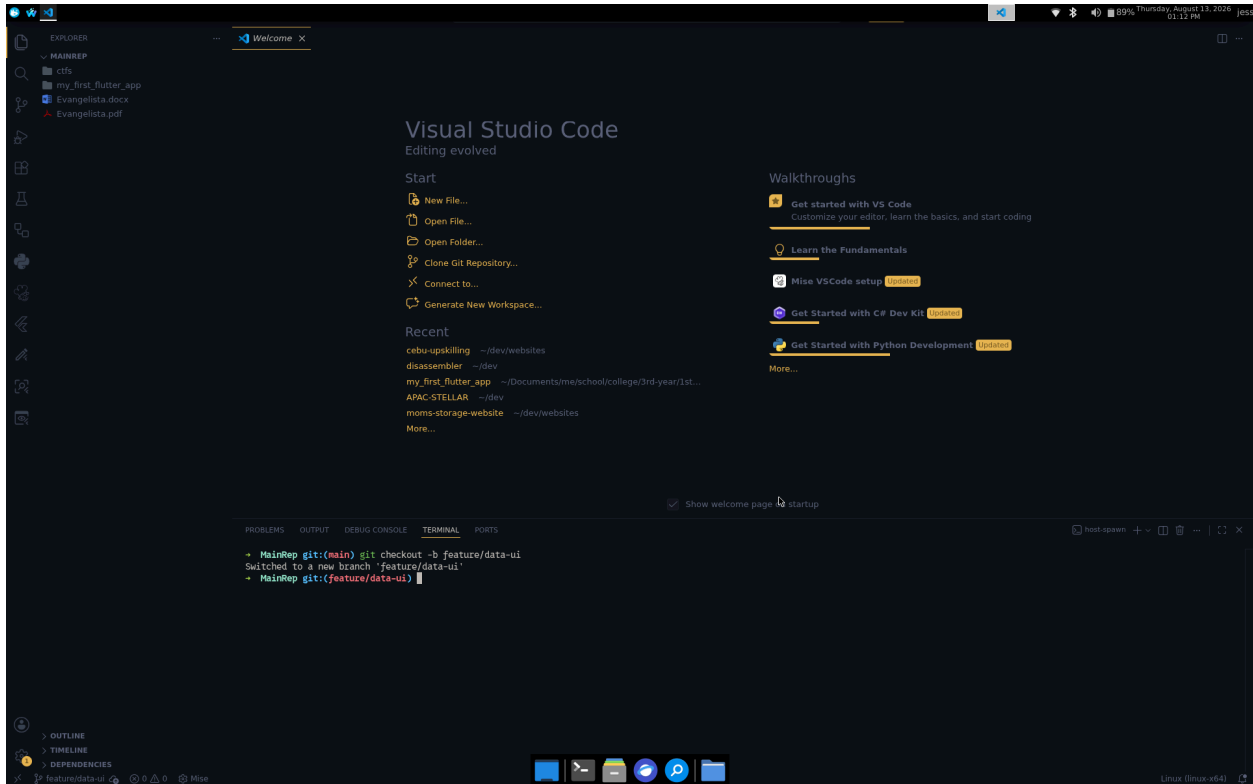
Objective: Create a feature branch before modifying your application.

Create:

`feature/data-ui`

Screenshot:

- Terminal showing your current branch



🚩 Flag 1 — Identify the Data

Objective: Identify the appropriate Dart data type for the information in your profile.

Research basic Dart data types and identify at least **four different types** that could be used in your application.

Examples:

String
int
double
bool

Create a table identifying the information and its appropriate data type.

Screenshot:

- Your identified data
- The corresponding data types



🚩 Flag 2 — Create Your Data

Objective: Create appropriately typed variables for your profile.

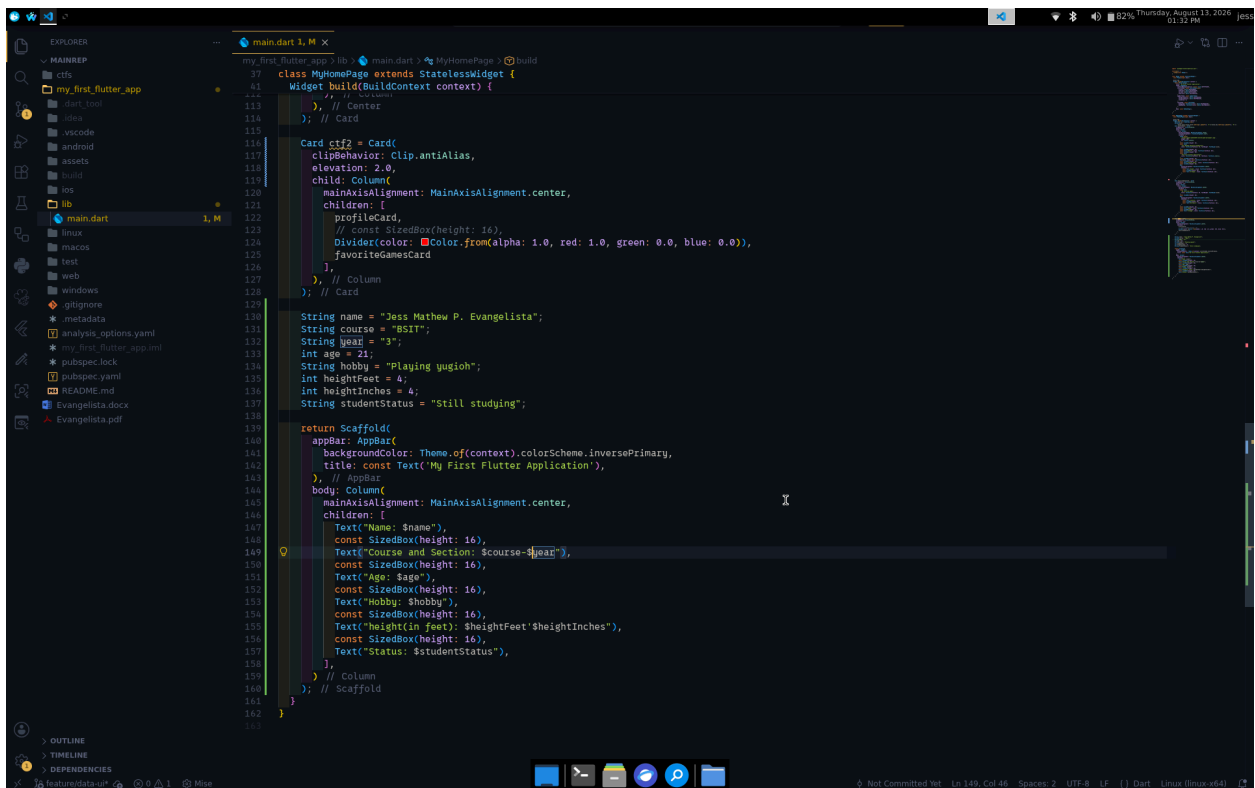
Create variables for at least:

- Name
- Course & Section
- Age
- Hobby
- Height
- Student status

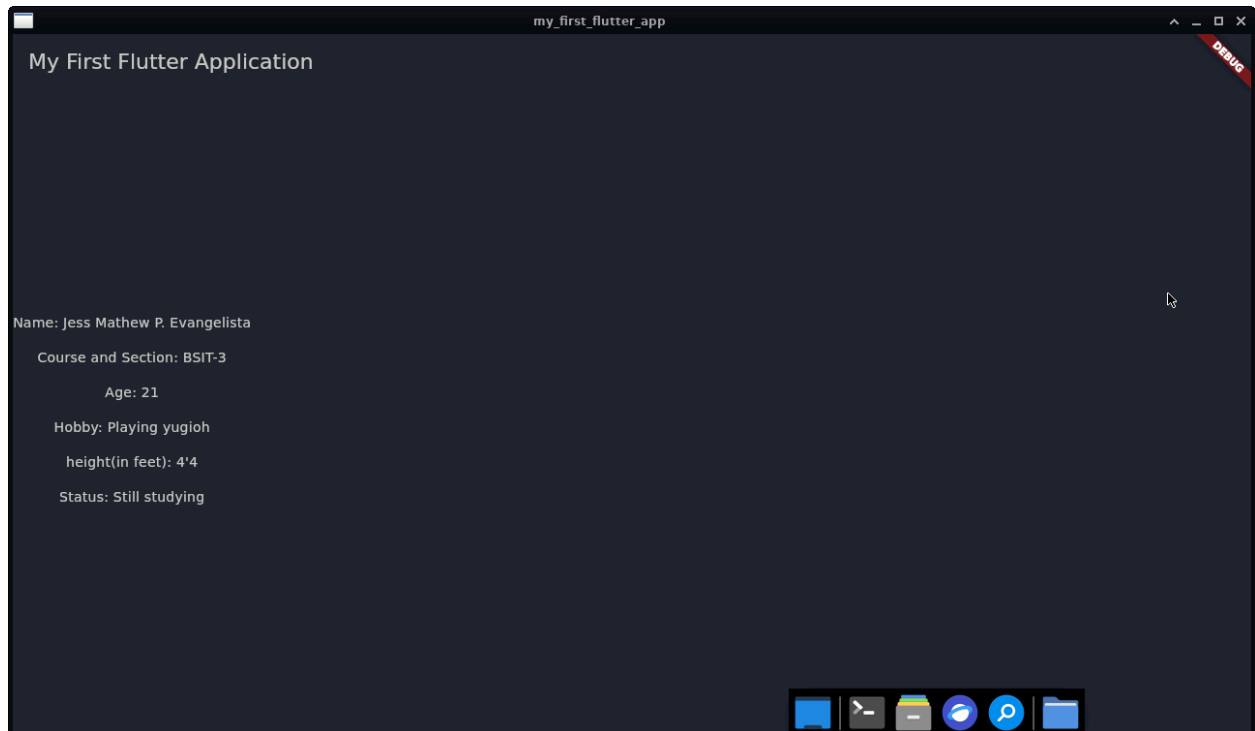
Your data should exist **separately** from your UI widgets.

Screenshot:

- Your variables
- Their data types
- Application still running



```
117 class MyHomePage extends StatelessWidget {
118   Widget build(BuildContext context) {
119     // Center
120     // Card
121     Card cf2 = Card(
122       clipBehavior: Clip.antiAlias,
123       elevation: 2.0,
124       child: Column(
125         mainAxisAlignment: MainAxisAlignment.center,
126         children: [
127           profileCard,
128           // const SizedBox(height: 16),
129           Divider(color: Color.from(alpha: 1.0, red: 1.0, green: 0.0, blue: 0.0)),
130           favoriteGamesCard
131         ],
132       ), // Column
133     ); // Card
134
135     String name = "Jess Mathew P. Evangelista";
136     String course = "RSIT";
137     String year = "3";
138     int age = 21;
139     String hobby = "Playing yugioh";
140     int heightFeet = 4;
141     int heightInches = 4;
142     String studentStatus = "Still studying";
143
144     return Scaffold(
145       appBar: AppBar(
146         backgroundColor: Theme.of(context).colorScheme.inversePrimary,
147         title: const Text('My First Flutter Application'),
148       ), // AppBar
149       body: Column(
150         mainAxisAlignment: MainAxisAlignment.center,
151         children: [
152           Text('Name: $name'),
153           const SizedBox(height: 16),
154           Text('Course and Section: $course-$year'),
155           const SizedBox(height: 16),
156           Text('Age: $age'),
157           const SizedBox(height: 16),
158           Text('Hobby: $hobby'),
159           const SizedBox(height: 16),
160           Text('height(in feet): $heightFeet $heightInches'),
161           const SizedBox(height: 16),
162           Text('Status: $studentStatus'),
163         ],
164       ), // Column
165     ); // Scaffold
166   }
167 }
```



Flag 3 — Connect Data to UI

Objective: Replace your hardcoded information with your variables.

Your profile should now display information using the variables you created.

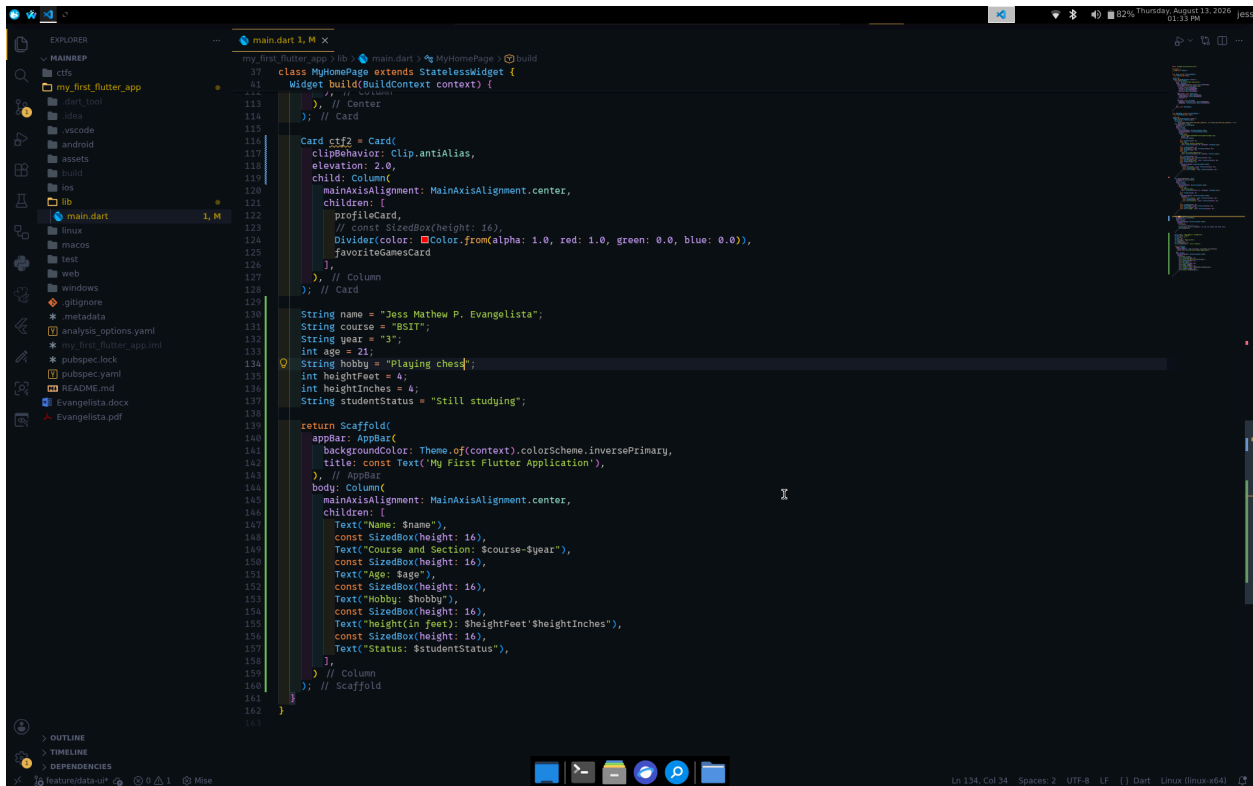
Test:

Change the value of one of your variables.

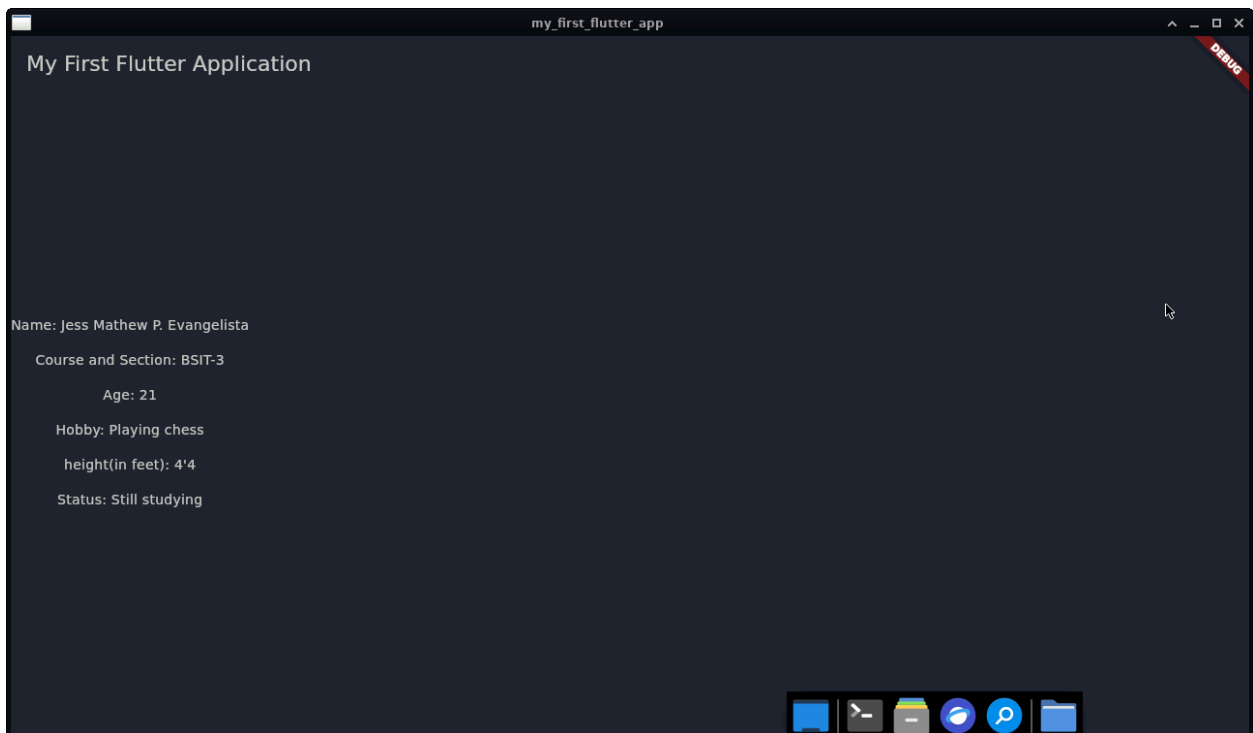
The UI should automatically display the new value.

Screenshot:

- Variables
- Widgets displaying the variables
- Updated application



```
37 class MyHomePage extends StatelessWidget {
38   Widget build(BuildContext context) {
39     // Center
40     // Card
41
42     Card ctf2 = Card(
43       clipBehavior: Clip.antiAlias,
44       elevation: 2.0,
45       child: Column(
46         mainAxisAlignment: MainAxisAlignment.center,
47         children: [
48           profileCard,
49           // const SizedBox(height: 16),
50           Divider(color: Color.from(alpha: 1.0, red: 1.0, green: 0.0, blue: 0.0)),
51           favoriteGamesCard
52         ],
53       ), // Column
54     ); // Card
55
56     String name = "Jess Mathew P. Evangelista";
57     String course = "BSIT";
58     String year = "3";
59     int age = 21;
60     String hobby = "Playing chess";
61     int heightFeet = 4;
62     int heightInches = 4;
63     String studentStatus = "Still studying";
64
65     return Scaffold(
66       appBar: AppBar(
67         backgroundColor: Theme.of(context).colorScheme.inversePrimary,
68         title: const Text('My First Flutter Application'),
69       ), // AppBar
70       body: Column(
71         mainAxisAlignment: MainAxisAlignment.center,
72         children: [
73           Text("Name: $name"),
74           const SizedBox(height: 16),
75           Text("Course and Section: $course-$year"),
76           const SizedBox(height: 16),
77           Text("Age: $age"),
78           const SizedBox(height: 16),
79           Text("Hobby: $hobby"),
80           const SizedBox(height: 16),
81           Text("height(in feet): $heightFeet'$heightInches'"),
82           const SizedBox(height: 16),
83           Text("Status: $studentStatus"),
84         ],
85       ), // Column
86     ); // Scaffold
87   }
88 }
```



🚩 Flag 4 — Make the Image Data-Driven

Objective: Make your profile image part of your data.

Create a variable containing the path to your profile image.

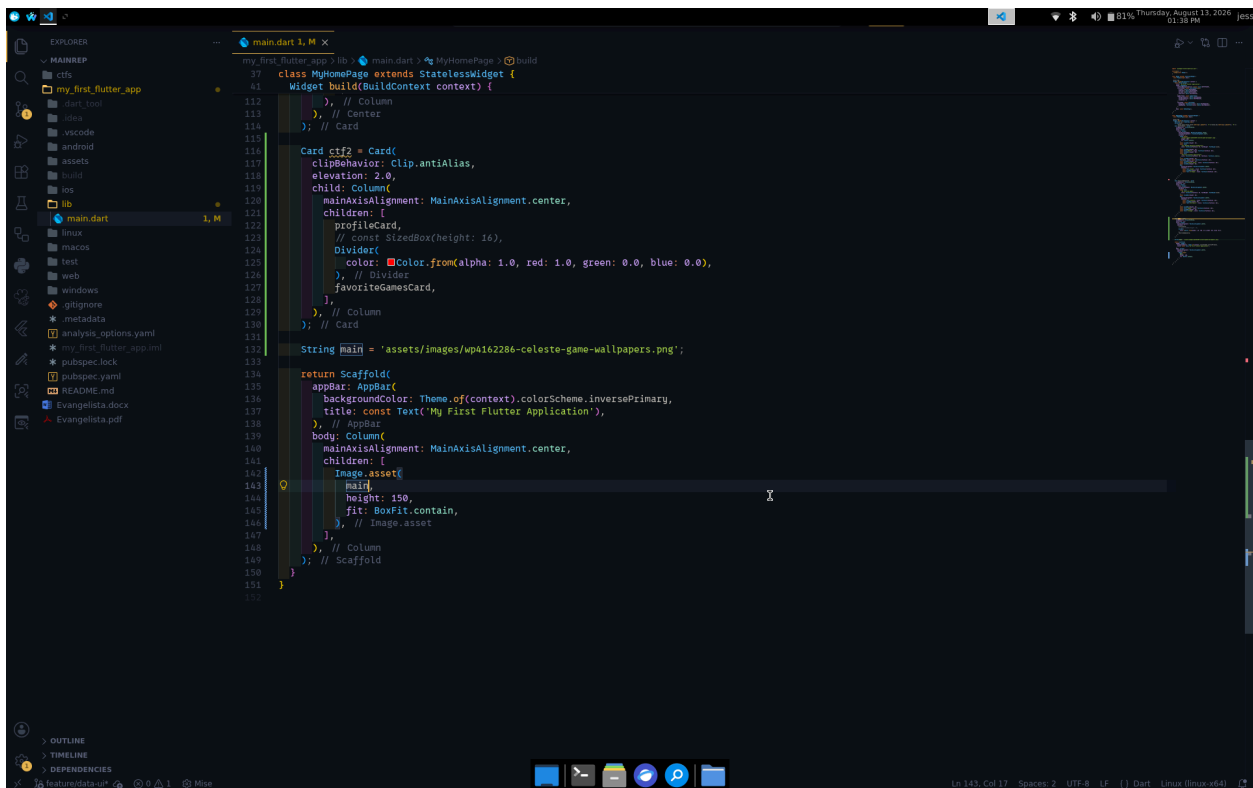
Use the variable when displaying the image.

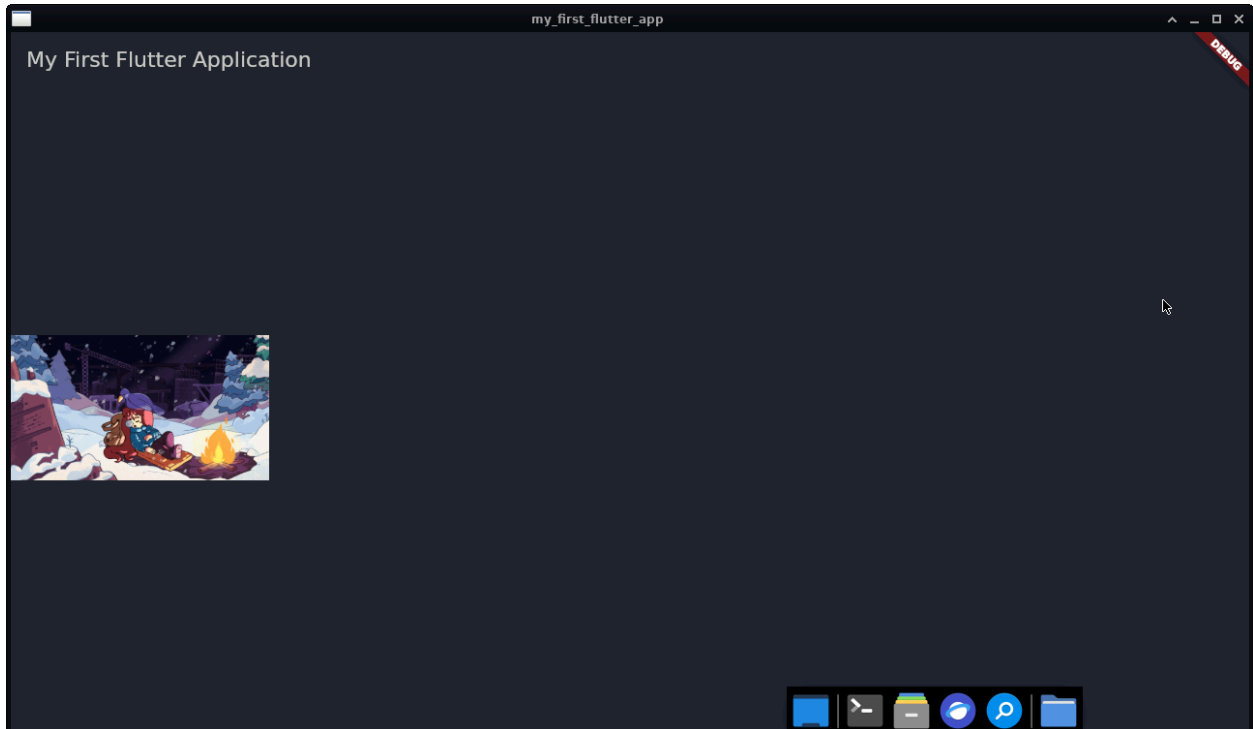
Then change the variable to another registered image.

The image displayed by the application should change.

Screenshot:

- Image variable
- Image widget using the variable
- Updated application





Flag 5 — Five Profiles, One Design

Objective: Create at least **five different profile cards** using the same visual design.

Each profile must contain at least:

- Image
- Name
- Course & Section
- Age
- Hobby

The five profiles must contain different information.

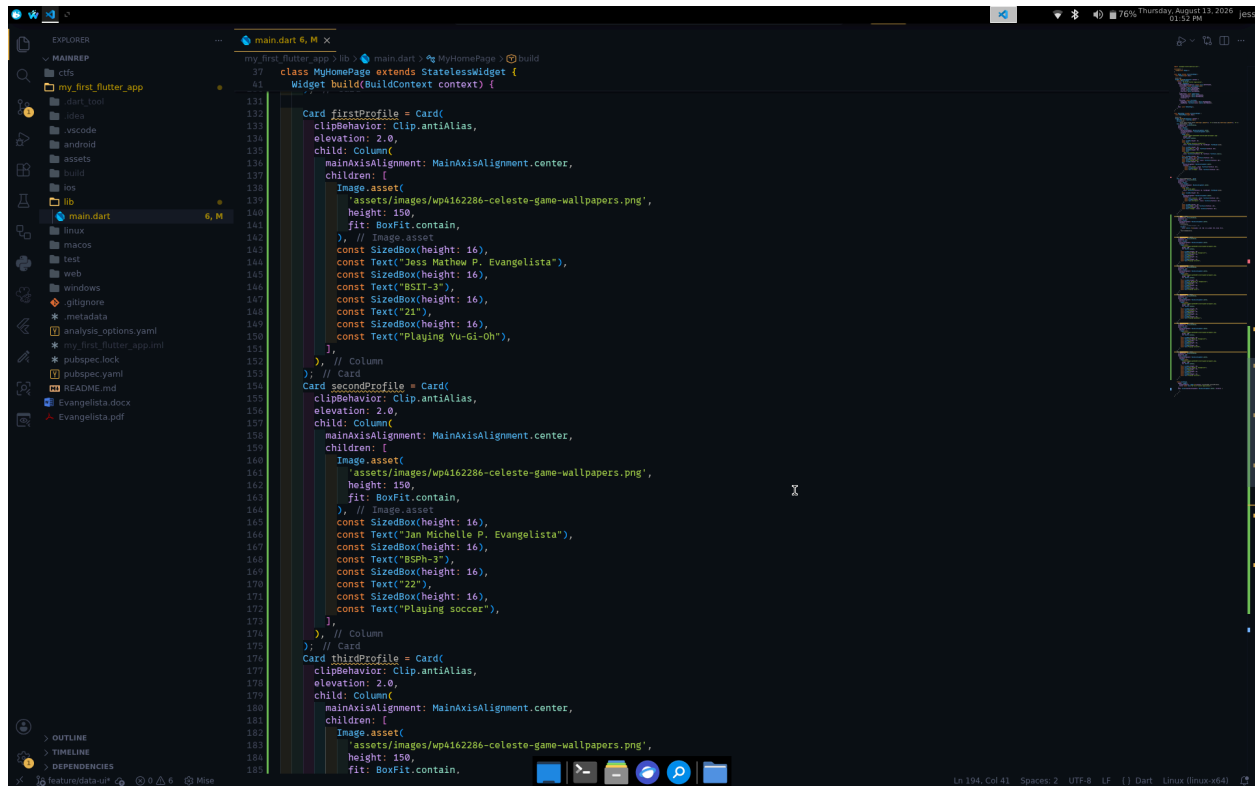
Important

All five profiles should use the **same visual design**.

You are demonstrating that the same UI can represent different data.

Screenshot:

- Your 5 different profile data



```
131 class MyHomePage extends StatelessWidget {
132   Widget build(BuildContext context) {
133     Card firstProfile = Card(
134       clipBehavior: Clip.antiAlias,
135       elevation: 2.0,
136       child: Column(
137         mainAxisAlignment: MainAxisAlignment.center,
138         children: [
139           Image.asset(
140             'assets/images/wp4162286-celeste-game-wallpapers.png',
141             height: 150,
142             fit: BoxFit.contain,
143           ), // Image.asset
144           const SizedBox(height: 16),
145           const Text("Jess Mathew P. Evangelista"),
146           const SizedBox(height: 16),
147           const Text("BSIT-3"),
148           const SizedBox(height: 16),
149           const Text("21"),
150           const SizedBox(height: 16),
151           const Text("Playing Yu-Gi-Oh"),
152         ], // Column
153       ); // Card
154     Card secondProfile = Card(
155       clipBehavior: Clip.antiAlias,
156       elevation: 2.0,
157       child: Column(
158         mainAxisAlignment: MainAxisAlignment.center,
159         children: [
160           Image.asset(
161             'assets/images/wp4162286-celeste-game-wallpapers.png',
162             height: 150,
163             fit: BoxFit.contain,
164           ), // Image.asset
165           const SizedBox(height: 16),
166           const Text("Jan Michelle P. Evangelista"),
167           const SizedBox(height: 16),
168           const Text("BSPh-3"),
169           const SizedBox(height: 16),
170           const Text("22"),
171           const SizedBox(height: 16),
172           const Text("Playing soccer"),
173         ], // Column
174       ); // Card
175     Card thirdProfile = Card(
176       clipBehavior: Clip.antiAlias,
177       elevation: 2.0,
178       child: Column(
179         mainAxisAlignment: MainAxisAlignment.center,
180         children: [
181           Image.asset(
182             'assets/images/wp4162286-celeste-game-wallpapers.png',
183             height: 150,
184             fit: BoxFit.contain,
```

🚩 Flag 6 — THE HIDDEN PROBLEM

You didn't see your UI?

Objective: Make all five profiles accessible to the user.

You just added five profile cards.

Now run your application.

Something is wrong.

Look carefully at your emulator.

Can you see all five profiles?

Maybe you can see the first one...

Maybe part of the second...

But where are the rest?

That's right.

You can't see them.

Your UI is larger than the available screen.

The profiles exist.

The data exists.

The widgets exist.

But the user can't access everything.

Your Mission:

Figure out how Flutter allows users to **scroll through content that is larger than the available screen**.

Research Flutter's **scrolling widgets** and determine which one is appropriate for your current layout.

You must make it possible to scroll through all five profiles.

Requirement:

The user must be able to:

Scroll → see Profile 1 → Profile 2 → Profile 3 → Profile 4 → Profile 5

without changing the size of the screen.

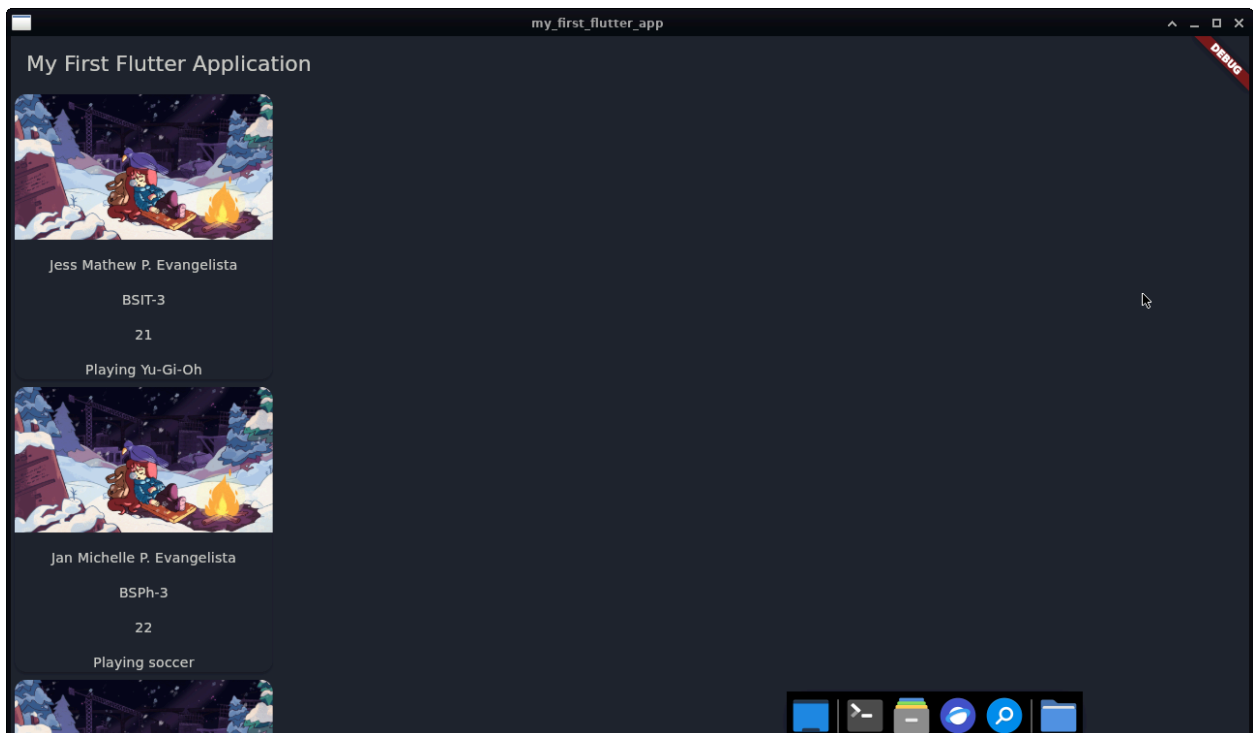
Screenshot:

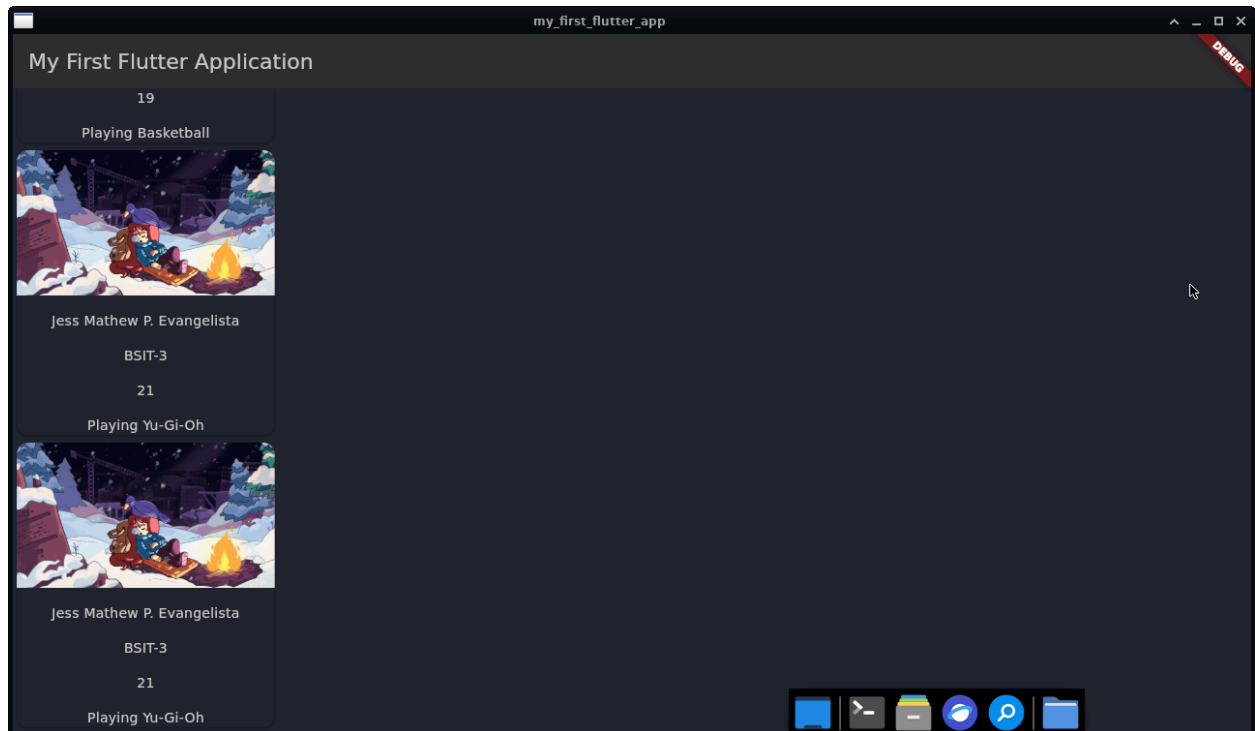
- The scrolling widget in your code
- Top portion of the application
- The rest of the portion of the application showing the later profiles

```
EXPLORER
  my_first_flutter_app
    main.dart
  lib
    main.dart
  linux
  macos
  test
  web
  windows
  .gitignore
  metadata
  analysis_options.yaml
  my_first_flutter_app.iml
  pubspec.lock
  pubspec.yaml
  README.md
  Evangelista.docx
  Evangelista.pdf

main.dart
class MyHomePage extends StatelessWidget {
  Widget build(BuildContext context) {
    child: Column(
      // mainAxisAlignment: MainAxisAlignment.center,
      children: [
        Image.asset(
          'assets/images/wp4162286-celeste-game-wallpapers.png',
          height: 150,
          fit: BoxFit.contain,
        ), // Image.asset
        const SizedBox(height: 16),
        const Text("Jess Mathew P. Evangelista"),
        const SizedBox(height: 16),
        const Text("BSIT-3"),
        const SizedBox(height: 16),
        const Text("21"),
        const SizedBox(height: 16),
        const Text("Playing Yu-Gi-Oh"),
      ],
    ), // Column
  ); // Card

  return Scaffold(
    appBar: AppBar(
      backgroundColor: Theme.of(context).colorScheme.inversePrimary,
      title: const Text('My First Flutter Application'),
    ), // AppBar
    body: SingleChildScrollView(
      child: Column(
        children: [
          firstProfile,
          secondProfile,
          thirdProfile,
          fourthProfile,
          fifthProfile,
        ],
      ), // Column
    ), // SingleChildScrollView
  ); // Scaffold
}
```





🚩 Flag 7 — Survive the Database

Objective: Prepare your UI for incomplete data.

Now imagine your profiles aren't manually created.

Someday, the data will come from a database.

And real data isn't always perfect.

Some information may be:

- Missing
- Empty
- `null`
- Not provided

Create at least **three profiles with missing information**.

For example:

Profile A

Name: Maria

Course: BSIT

Hobby: missing

Profile B

Name: Juan

Course: missing

Hobby: Gaming

Profile C

Name: missing

Course: MMA

Hobby: Drawing

Your application must **not crash** when information is missing.

Instead, provide an appropriate fallback.

Examples:

Hobby: Not provided

Course: Unknown

Challenge

Your placeholders should look intentional and clean.

Screenshot:

- Incomplete profile data
- Application displaying fallback values
- Application still functioning correctly

```
37 class MyHomePage extends StatelessWidget {
38   Widget build(BuildContext context) {
39     // This widget is the root of your application.
40     return Scaffold(
41       appBar: AppBar(
42         title: const Text('My First Flutter Application'),
43       ),
44       body: ListView.builder(
45         itemCount: profiles['names'].length,
46         itemBuilder: (ctx, ind) {
47           return Column(
48             children: [
49               Text(profiles['names'][ind]??.toString() ?? "Not provided"),
50               const SizedBox(height: 16),
51             ],
52           );
53         },
54       ),
55     );
56   }
57 }
58
59 var profiles = {
60   "names": ["Maria", "Jess", "Jan", "Jake"],
61   "courseSection": ["BSIT-3", null, "BSIT-3", "BSIT-3"],
62   "age": [21, 22, null, 23, 21],
63   "hobby": ["Gaming", "Basketball", "Soccer", null, "Reading"],
64 };
65
66 return Scaffold(
67   appBar: AppBar(
68     backgroundColor: Theme.of(context).colorScheme.inversePrimary,
69     title: const Text('My First Flutter Application'),
70   ),
71   // AppBar
72   body: ListView.builder(
73     itemCount: profiles['names'].length,
74     itemBuilder: (ctx, ind) {
75       return Column(
76         children: [
77           Text(profiles['names'][ind]??.toString() ?? "Not provided"),
78           const SizedBox(height: 16),
79         ],
80       );
81     },
82   ),
83 );
```



🚩 Flag 8 — Save Your Progress

Objective: Commit and push your completed Data-Driven UI CTF.

Use an appropriate commit message following proper Git standards.

Branch:

`feature/data-ui`

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit